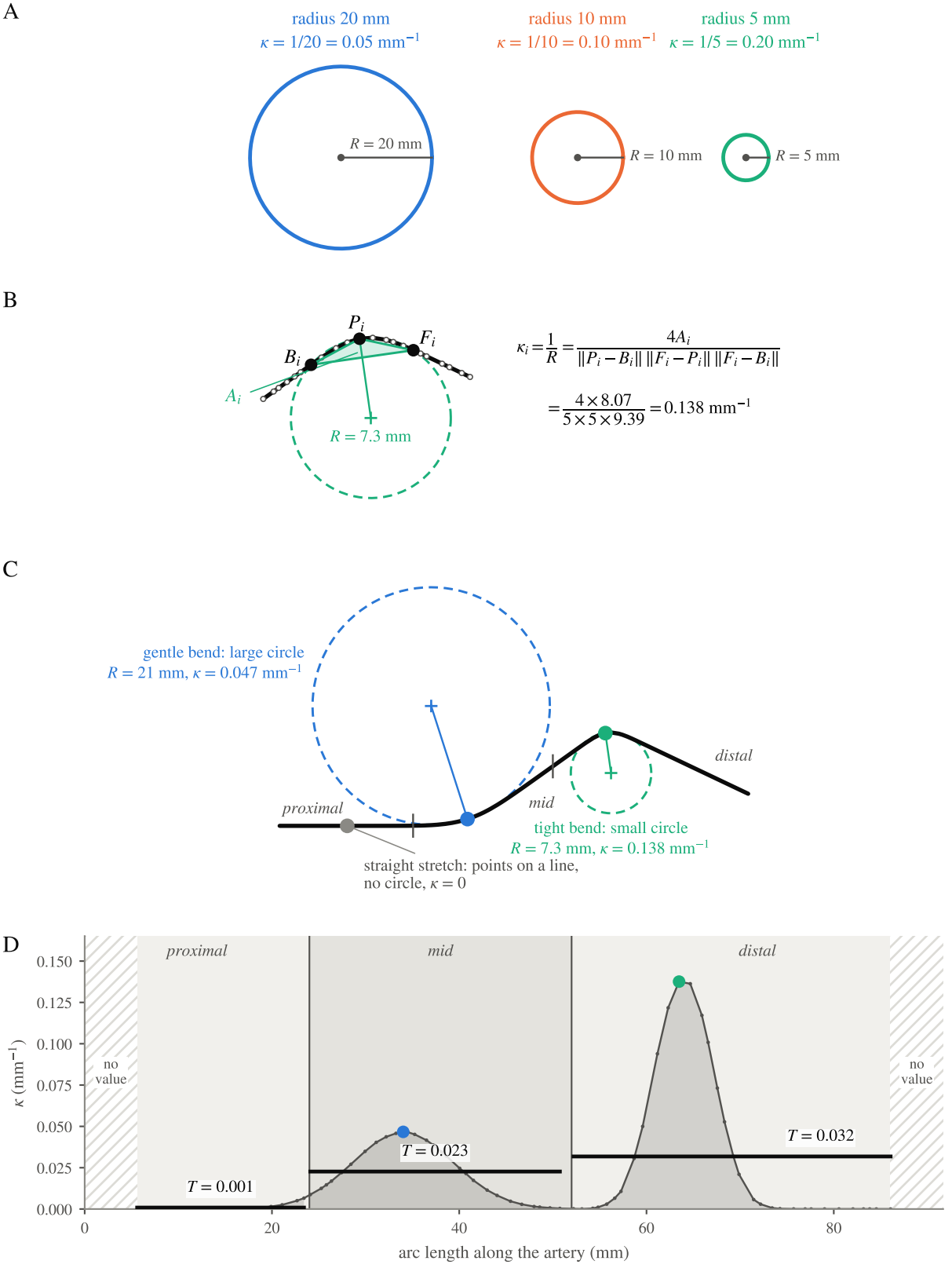
A
radius 20 mm
$\kappa = 1/20 = 0.05$ mm^{-1}
radius 10 mm
$\kappa = 1/10 = 0.10$ mm^{-1}
radius 5 mm
$\kappa = 1/5 = 0.20$ mm^{-1}
$R = 20$ mm
$R = 10$ mm
$R = 5$ mm
B
P_i
B_i
F_i
A_i
$R = 7.3$ mm
$\kappa_i = \frac{1}{R} = \frac{4A_i}{\|P_i - B_i\| \, \|F_i - P_i\| \, \|F_i - B_i\|}$
$= \frac{4 \times 8.07}{5 \times 5 \times 9.39} = 0.138$ mm^{-1}
C
gentle bend: large circle
$R = 21$ mm, $\kappa = 0.047$ mm^{-1}
proximal
mid
distal
tight bend: small circle
$R = 7.3$ mm, $\kappa = 0.138$ mm^{-1}
straight stretch: points on a line,
no circle, $\kappa = 0$
D
proximal
mid
distal
no value
no value
$T = 0.001$
$T = 0.023$
$T = 0.032$
κ (mm^{-1})
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
20
40
60
80
arc length along the artery (mm)